

Project: Customer and Banking Transaction Behaviour Analysis

A. Business Context

Business Problem: A retail bank wants to better understand how different customer groups conduct transactions, identify high-activity and under-engaged segments, and evaluate the adoption of digital transaction channels. The analysis aims to support customer engagement and channel strategy using transaction and customer profile data.

Business Objective: Analyze customer transaction behavior across segments, channels and demographic groups to identify actionable opportunities for customer engagement and digital-channel development.

Stakeholders: Customer Strategy, Digital Banking and Branch Operations,...

Business Questions:

1. How do different customer segments behave in terms of transaction frequency and value?
2. Which customer segments show the highest and lowest transaction activity?
3. Which transaction channels are most commonly used by each customer segment, and how strong is Mobile App adoption?
4. How does transaction behavior differ across age groups and genders?
5. How does channel usage vary across different times of the day?
6. Who are the highest-value customers based on total transaction value?

Tools: Python, Pandas, Matplotlib, Data Visualization, Exploratory Data Analysis

B. Data Overview and Preparation

Data Source: Vietnamese Bank from 2025 January 1st to 2025 July 1st

Data Overview: as in the figure 1.

- The dataset "Transaction" includes TransactionID, CustomerID, Date, Time, Transaction Amount, Location, Channel, Status,...
- The Dataset "Customer" includes CustomerID, Name, Gender, Age, Occupation, Customer Segment, Contact Information,....

Data Preparation:

Customer and transaction datasets were joined using Customer ID (MaKH) to enable behavioral analysis across customer profiles, segments and transaction channels.

Load Dataset

Due to the limitation in the storage, I cannot upload the whole dataset here. Please contact for details

```
import pandas as pd
```

```
df_gd = pd.read_csv("/content/giao_dich.csv")  
df_kh = pd.read_csv("/content/khach_hang_crm.csv")
```

```
df_gd.head(5)
```

	MaGD	MaKH	NgayGD	GioGD	LoaiGD	SoTien	ChiNhanh	KenhGD	TrangThai	GhiChu
0	GD000508	KH0124	2024-02-17	16:16	Mua ngoại tệ	165717229	Hà Nội	Mobile App	Thành công	NaN
1	GD000082	KH0198	2024-01-08	16:04	Nạp tiền	20738329	Khánh Hòa	ATM	Thành công	NaN
2	GD000091	KH0033	2024-01-24	13:07	Chuyển khoản	154100866	Đồng Nai	Internet Banking	Thành công	NaN
3	GD000193	KH0095	2024-06-30	16:02	Chuyển khoản	58005231	Nghệ An	POS	Thành công	NaN
4	GD000474	KH0133	2024-06-02	17:27	Rút tiền	165478158	Đồng Nai	POS	Thành công	NaN

```
df_kh.head(5)
```

	MaKH	HoTen	GioiTinh	Tuoi	NgheNghiep	NhomKH	Email	SoDienThoai
0	KH0001	Phạm Văn Ích	Nam	34	Giáo viên	Cá nhân thường	kh0001@email.com	983197857
1	KH0002	Lê Bảo Oanh	Nam	21	Kinh doanh tự do	Cá nhân VIP	kh0002@email.com	941227216
2	KH0003	Tạ Bảo An	Nam	65	Luật sư	Doanh nghiệp lớn	kh0003@email.com	939587039
3	KH0004	Dương Bảo Ích	Nam	68	Giáo viên	Doanh nghiệp lớn	kh0004@email.com	955667651
4	KH0005	Võ Hoàng Giang	Nữ	26	Kinh doanh tự do	Doanh nghiệp lớn	kh0005@email.com	922981052

```
df=pd.merge(df_gd,df_kh, on="MaKH", how="left")  
df.head(5)
```

...	MaGD	MaKH	NgayGD	GioGD	LoaiGD	SoTien	ChiNhanh	KenhGD	TrangThai	GhiChu	HoTen	GioiTinh	Tuoi
0	GD000508	KH0124	2024-02-17	16:16	Mua ngoại tệ	165717229	Hà Nội	Mobile App	Thành công	NaN	Mai Thủy Nam	Nữ	40
1	GD000082	KH0198	2024-01-08	16:04	Nạp tiền	20738329	Khánh Hòa	ATM	Thành công	NaN	Lê Thanh Tâm	Nữ	61
2	GD000091	KH0033	2024-01-24	13:07	Chuyển khoản	154100866	Đồng Nai	Internet Banking	Thành công	NaN	Dương Thanh Oanh	Nữ	29

Figure 1. Data Samples

Data Preprocessing: Remove the missing values/duplicates

C. Customer Segment Performance

Analysis Coding

Key Findings: Large corporate customers show the highest transaction count and total transaction value in the sample, while small corporate customers show comparatively lower activity.

Business Interpretation: Large corporate customers show the highest transaction count and total transaction value in the sample, while small corporate customers show comparatively lower activity.

Recommended Action: Investigate the transaction needs and product usage of small corporate customers before designing targeted engagement initiatives.

D. Transaction Channel and Digital Adoption

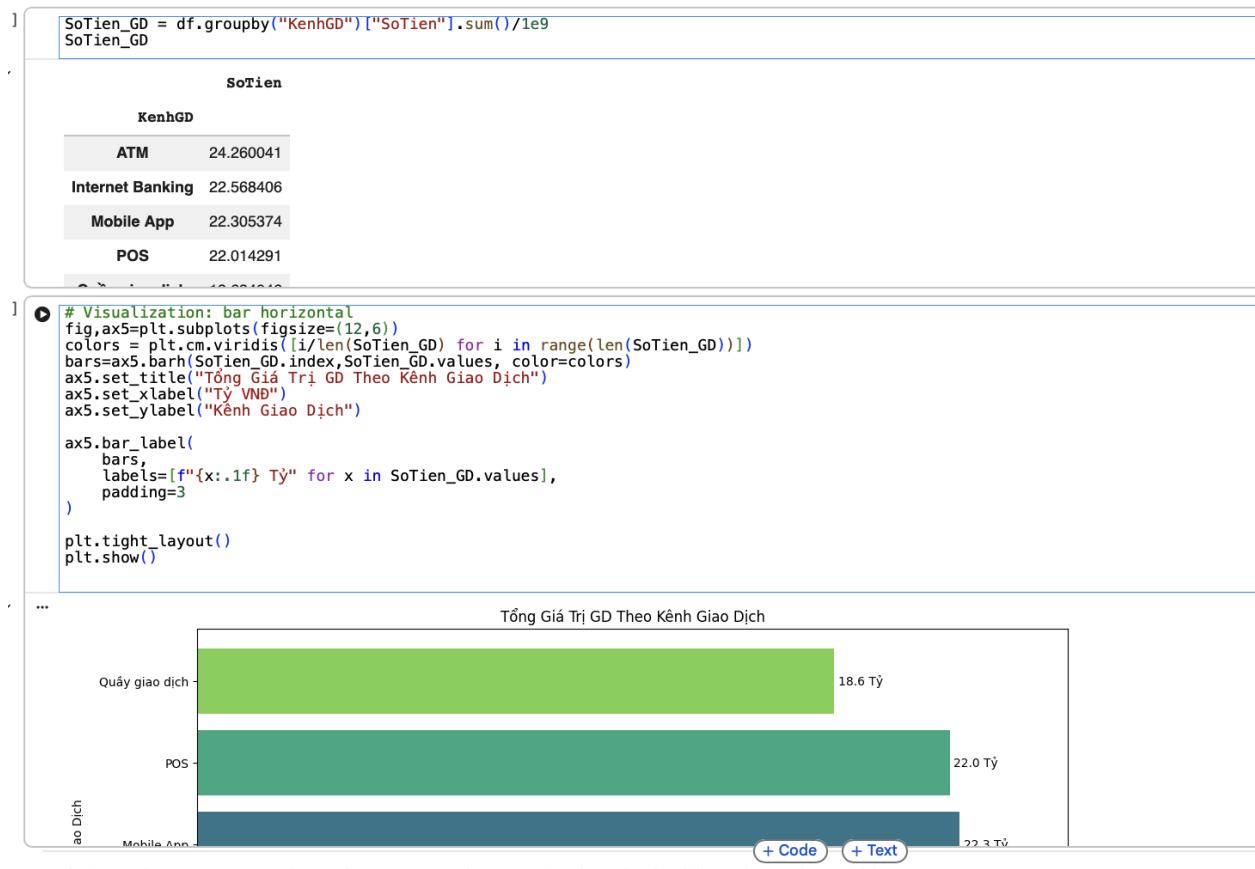
- Analysis Coding:

3. Which transaction channels are most popular in each customer segment? Has the Mobile App been widely adopted?

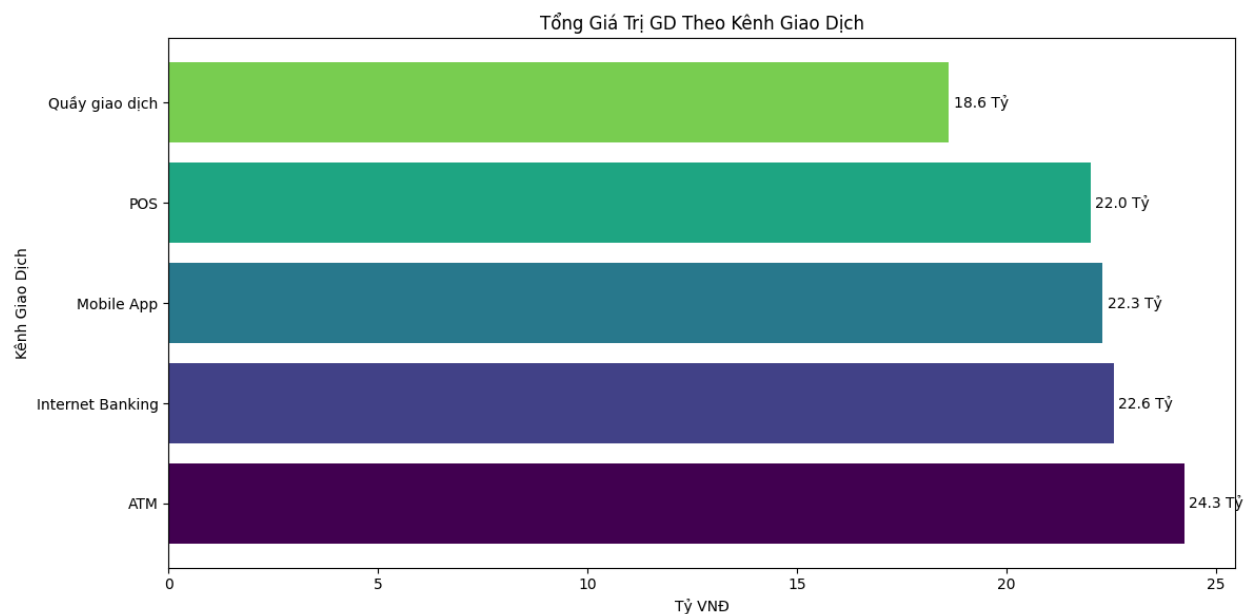
```
df.groupby(["NhómKH"], ["KênhGD"]).value_counts()
```

	NhómKH	KênhGD	count
	Cá nhân VIP	Mobile App	57
		Internet Banking	53
		ATM	48
		POS	44
		Quầy giao dịch	37
	Cá nhân thường	Internet Banking	53
		POS	53
		ATM	48
		Mobile App	46
		Quầy giao dịch	40
	Doanh nghiệp lớn	ATM	56
		POS	55
		Internet Banking	54
		Quầy giao dịch	49
		Mobile App	47
	Doanh nghiệp nhỏ	Quầy giao dịch	60
		Mobile App	54
		ATM	45
		POS	40
		Internet Banking	38

dtype: int64



- **Visualization:**



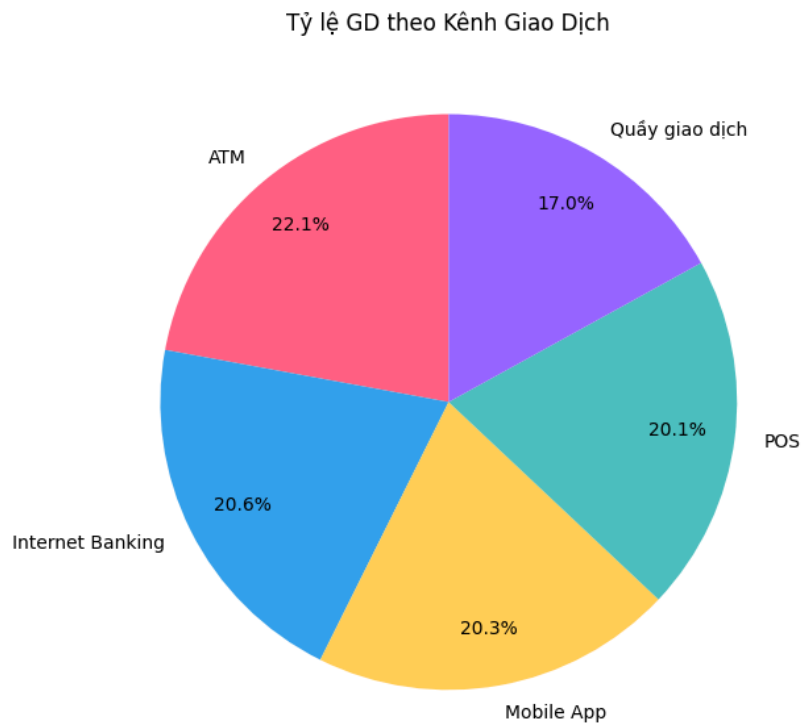


Figure 3. Coding and Visualization for Transaction Channel and Digital Adoption

Key Findings: Mobile App is not consistently the most frequently used channel across customer segments, suggesting that digital adoption varies by customer group.

Business Interpretation: Channel strategies should therefore be differentiated by customer segment rather than assuming one preferred channel for all customers.

Recommended Action: Identify segments with relatively low Mobile App usage and investigate whether adoption is constrained by awareness, customer needs, usability, or preference for other channels.

E. Demographic Transactions Behavior

Total Transaction Value by Age Group

- Analysis Coding

Define the age group

```
def phan_loai_tuoi(tuoiKH):  
    if tuoiKH >= 66:  
        return 'Senior(66+)'  
    elif 51 <= tuoiKH <= 65:  
        return 'Baby Boomers(51-65)'  
    elif 26 <= tuoiKH <= 35:  
        return 'Millennials(26-35)'  
    elif 36 <= tuoiKH <= 50:  
        return 'Gen X(36-50)'  
    else:  
        return 'Gen Z(18-25)'
```

```
df["NhomTuoi"] = df["Tuoi"].apply(phan_loai_tuoi)  
df.head(5)
```

	MaGD	MaKH	NgayGD	GioGD	LoaiGD	SoTien	ChiNhanh	KenhGD	TrangThai	ChiChu	HoTen	GioiTinh	Tuoi	NghaNghiep	NhomKH	Email	SoDienThoai	NhomTuoi
0	GD000508	KH0124	2024-02-17	16:16	Mua ngoại tệ	165717229	Hà Nội	Mobile App	Thành công	NaN	Mai Thủy Nam	Nữ	40	Bác sĩ	Cá nhân VIP	kh0124@email.com	986745380	Gen X(36-50)
1	GD000082	KH0198	2024-01-08	16:04	Nạp tiền	20738329	Khánh Hòa	ATM	Thành công	NaN	Lê Thanh Tâm	Nữ	61	Sinh viên	Cá nhân VIP	kh0198@email.com	999941500	Baby Boomers(51-65)
2	GD000091	KH0033	2024-01-24	13:07	Chuyển khoản	154100866	Đồng Nai	Internet Banking	Thành công	NaN	Dương Thanh Oanh	Nữ	29	Bác sĩ	Doanh nghiệp nhỏ	kh0033@email.com	939219319	Millennials(26-35)
3	GD000193	KH0095	2024-06-30	16:02	Chuyển khoản	58005231	Nghệ An	POS	Thành công	NaN	Phạm Thủy Phúc	Nam	33	Nội trợ	Doanh nghiệp nhỏ	kh0095@email.com	989133014	Millennials(26-35)
4	GD000474	KH0133	2024-06-02	17:27	Rút tiền	165478158	Đồng Nai	POS	Thành công	NaN	Huỳnh Hoàng Phúc	Nam	59	Hưu trí	Cá nhân VIP	kh0133@email.com	976795767	Baby Boomers(51-65)

```
#Number of transactions for each age group  
SoLuongGD_NhomTuoi=df.groupby("NhomTuoi")["MaGD"].count()  
  
#Total volume of transactions for each age group  
TongTien_NhomTuoi =df.groupby("NhomTuoi")["SoTien"].sum()  
  
#Average volume of transaction for each age group  
GDTB_NhomTuoi = df.groupby("NhomTuoi")["SoTien"].mean()  
  
#Number of customer for each age group  
SoLuongKH_NhomTuoi = df.groupby("NhomTuoi")["MaKH"].nunique()  
  
#Transaction patterns across different age groups  
df_tonghop1= pd.DataFrame({  
    "Số GD": SoLuongGD_NhomTuoi,  
    "Tổng GD (Tỷ)": TongTien_NhomTuoi/1e9,  
    "GD TB (TR)": GDTB_NhomTuoi/1e6,  
    "Số KH": SoLuongKH_NhomTuoi  
})  
df_tonghop1
```

	Số GD	Tổng GD (Tỷ)	GD TB (TR)	Số KH
NhomTuoi				
Baby Boomers(51-65)	287	33.040381	115.123278	54
Gen X(36-50)	244	26.601661	109.023200	52
Gen Z(18-25)	148	15.851677	105.754575	27
Millennials(26-35)	238	26.950807	113.238685	44
Senior(66+)	103	12.063992	117.126136	21

● Visualization

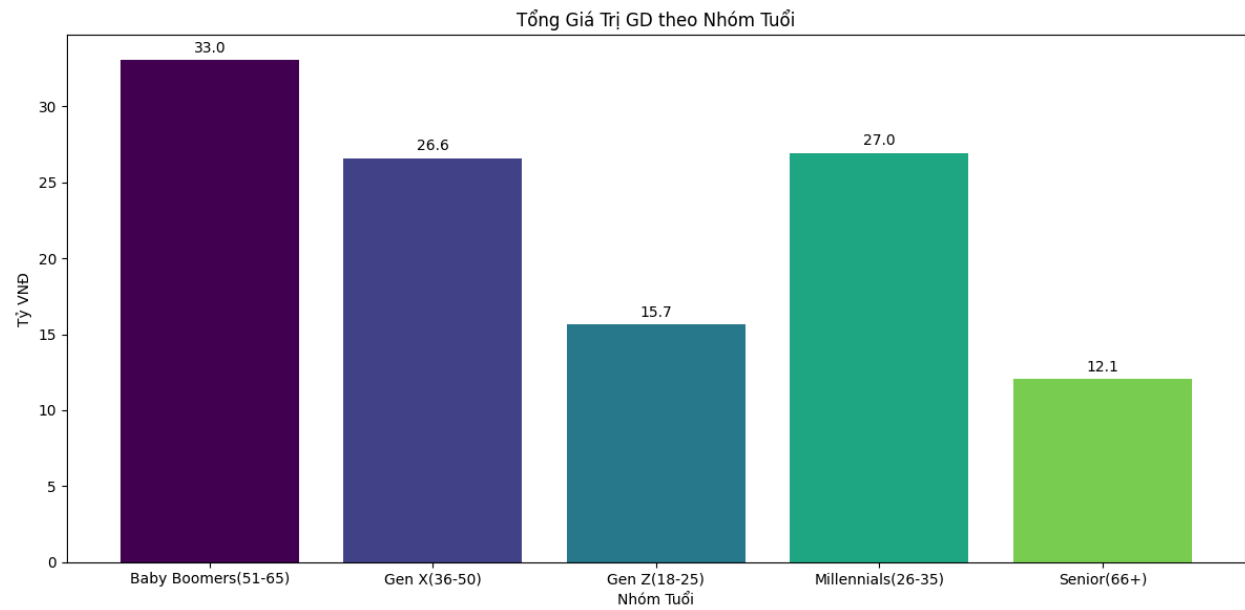


Figure 4. Coding and Visualization for Total Transaction Value by Age Group

Total Transaction Value by Gender

● Analysis Coding

```
#Transaction patterns across genders
TongTien_GioiTinh = (df.groupby("GioiTinh")["SoTien"].sum())/1e9
TongTien_GioiTinh
```

```
SoTien

GioiTinh
Nam      68.900297
Nữ       45.408221
```

```
dtype: float64
```

```
TongTien_GioiTinh.index
```

```
Index(['Nam', 'Nữ'], dtype='object', name='GioiTinh')
```

```
#Bar Chart
fig,ax2=plt.subplots(figsize=(12,6))
ax2.set_title("Tổng giá trị GD theo giới tính")
ax2.set_xlabel("Giới Tính")
ax2.set_ylabel("Tổng GD (Tỷ)")

colors = plt.cm.viridis([i/len(TongTien_GioiTinh) for i in range(len(TongTien_GioiTinh))])
bars=ax2.bar(TongTien_GioiTinh.index,TongTien_GioiTinh.values, color=colors)
ax2.bar_label(
    bars,
    labels = [f"{x:.1f} Tỷ" for x in TongTien_GioiTinh.values],
    padding=3
)

plt.tight_layout()
plt.show()
```

- *Visualization*

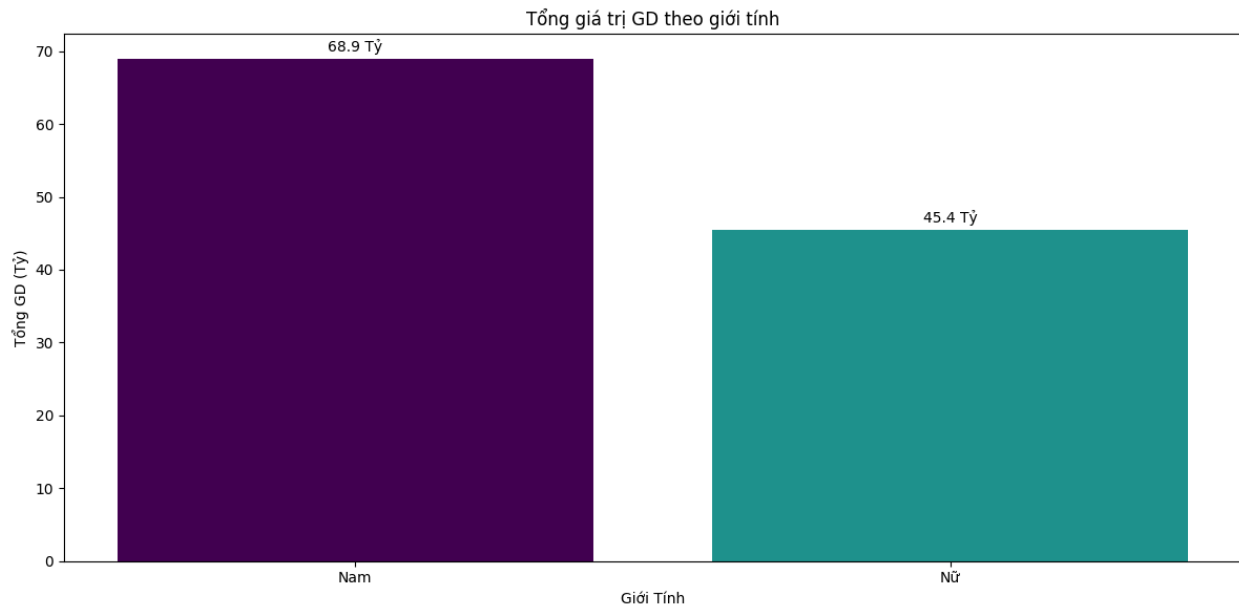


Figure 4. Coding and Visualization for Total Transaction Value by Genders

Key Findings: Baby boomers contribute the highest number and largest volume of transactions among 5 provided age groups while males have higher volume of transactions than females

F. Channel Preferences by Age Group

- Analysis Coding

```
1 import pandas as pd

1 pivot_heat = pd.pivot_table(
    df,
    index = "NhomTuoi",
    values = "MaGD",
    columns = "KenhGD",
    aggfunc = "count",
    fill_value = 0
)

1 pivot_heat
```

	KenhGD	ATM	Internet Banking	Mobile App	POS	Quầy giao dịch
NhomTuoi						
Baby Boomers(51-65)		48	56	51	64	54
Gen X(36-50)		45	57	53	40	39
Gen Z(18-25)		30	30	36	22	25
Millennials(26-35)		46	36	48	42	54
Senior(66+)		28	19	16	24	14

```
1 fig,ax4=plt.subplots(figsize=(12,6))
  sns.heatmap(pivot_heat, cmap="YlGnBu", annot=True, fmt=".0f", ax=ax4)
  ax4.set_title("Heatmap: Số GD Theo Nhóm Tuổi x Kênh GD")
  ax4.set_xlabel("Kênh Giao Dịch")
  ax4.set_ylabel("Nhóm Tuổi")

  plt.tight_layout()
  plt.show()
```

- Visualization

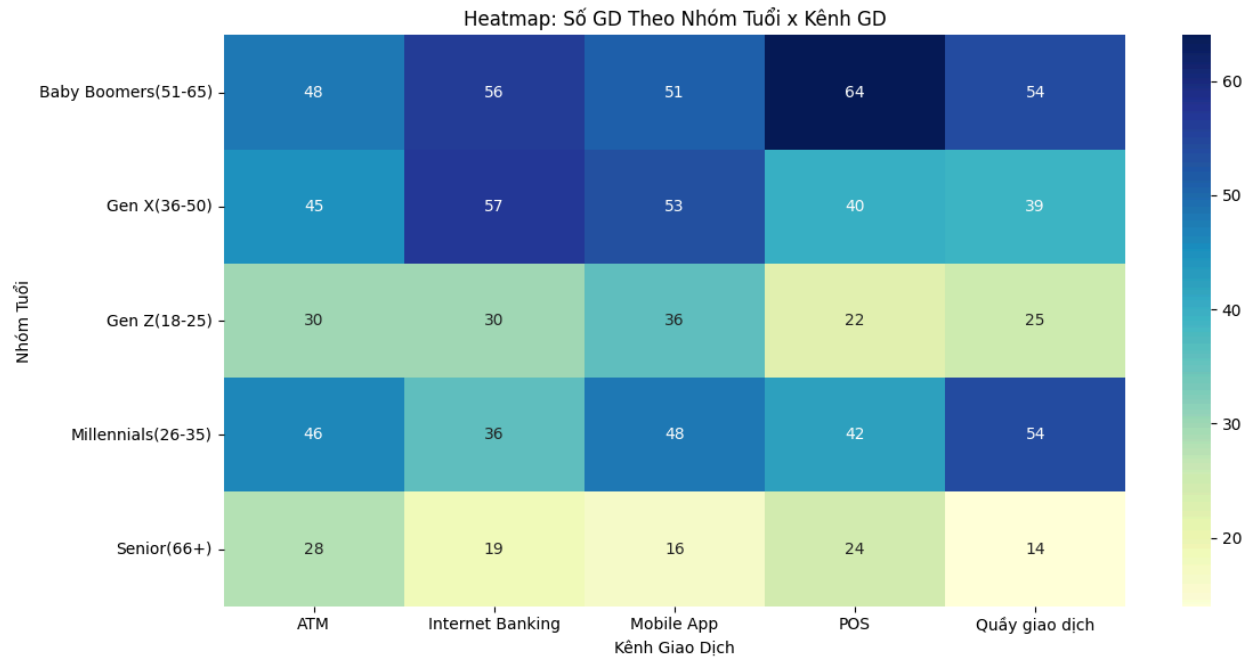


Figure 5. Coding and Visualization for Channel Preferences by Age Group

Key Finding: Transaction-channel usage differs across age groups, indicating that channel preferences are not uniform across the customer base.

Business Implication: Customer communication and digital-adoption strategies may be more effective when tailored by age group.

G. Time-of-Day Analysis by Channel

- *Analysis Coding*
- *Visualization*

Key Findings:

- For ATM, 2pm is the peak time for transactions
- For Internet Banking, 4pm is the peak time for transactions
- For Mobile App, 3pm is the peak time for transactions
- For POS, 1pm is the peak time for transactions
- For offline banking, 9am is the peak time for transactions

Business Implications: Based on the peak transaction time by channel, marketing managers and customer services teams can plan the effective marketing messages and also support digital-service monitoring, staffing.

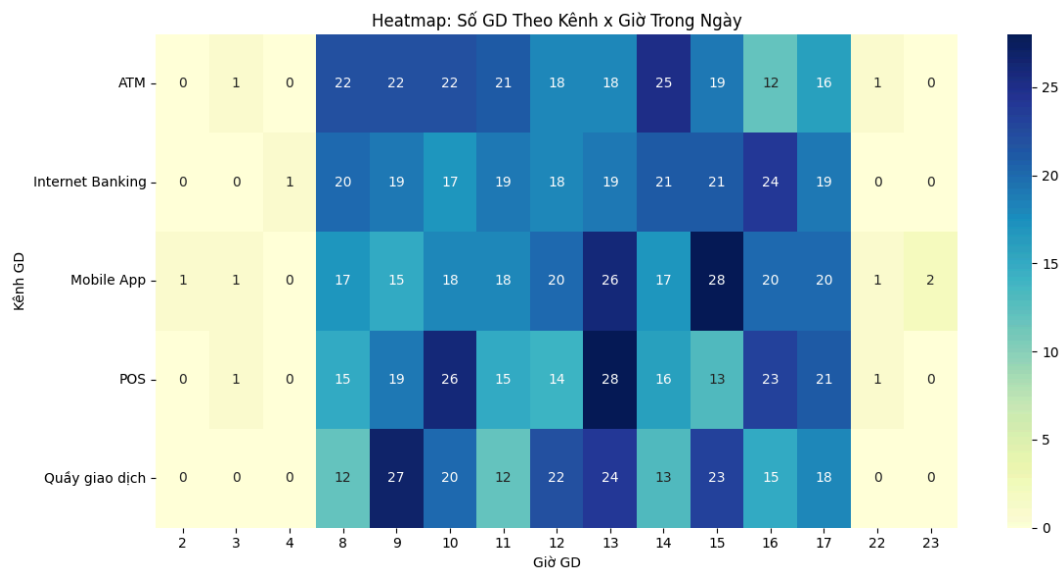
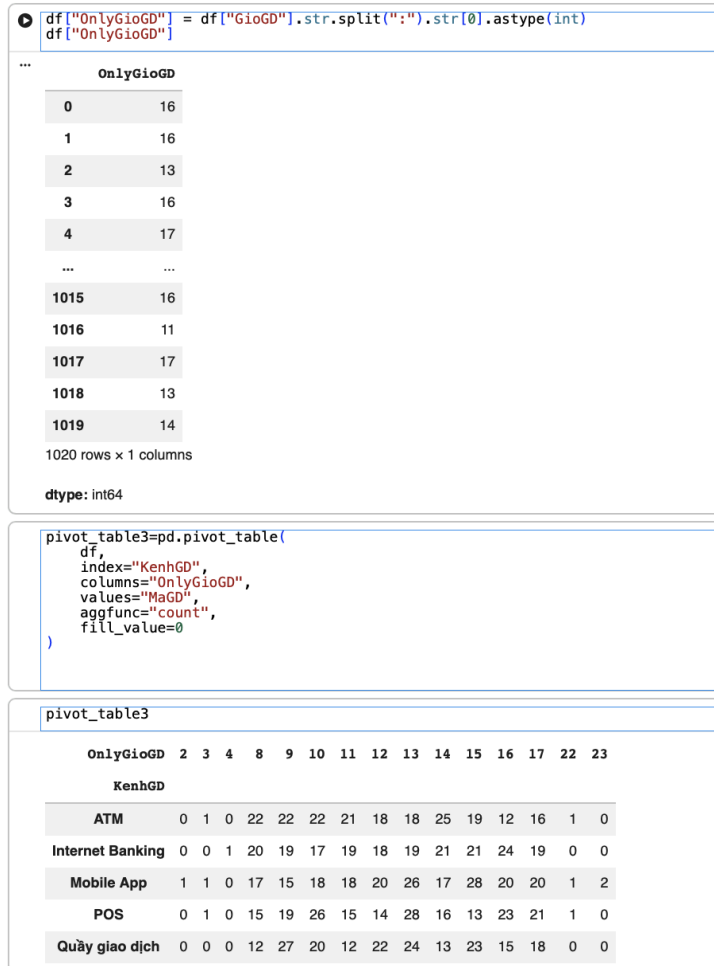


Figure 5. Coding and Visualization for Time-of-day analysis by channel

H. Top 20 High-Value Customers

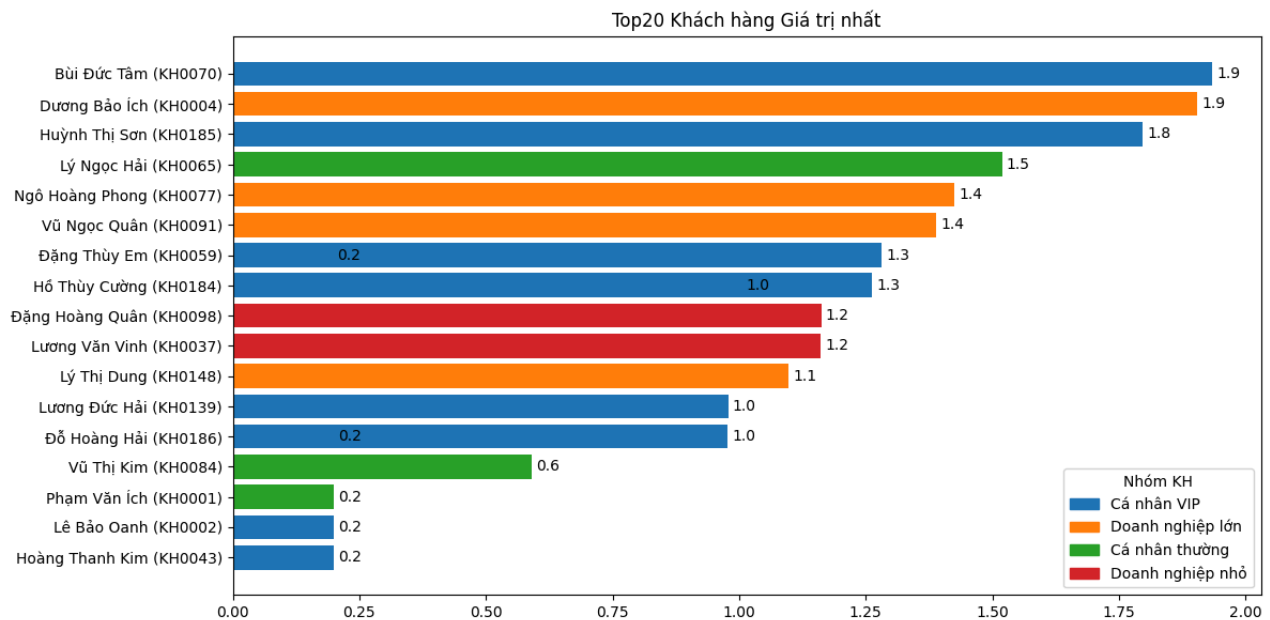


Figure 5. Visualization for Top 20 High-Value Customers

Business Interpretation: A relatively small group of customers may account for a meaningful share of total transaction value, making them important candidates for retention and relationship-management efforts.

I. Executive Summary

- *Customer Segment Strategy:* Large corporate customers currently show the highest overall transaction activity, while small corporate customers warrant further investigation to understand their lower activity.
- *Digital Channel Strategy:* Mobile App usage varies across customer segments and is not universally dominant. Digital-adoption efforts should therefore focus on segments with comparatively low usage.
- *Customer Engagement:* Demographic and channel-preference differences suggest that a single customer-engagement strategy may not suit all customer groups.
- *High-Value Customer Management:* Customers with high cumulative transaction value should be monitored separately to support retention and relationship-management initiatives.

Note: Please find the full coding file attached